

Matrix Demonstrations

Estimation I

Throughout, $\mathbf{A}$ and $\mathbf{B}$ are conformable and nonsingular unless otherwise noted.

Inverse Properties

Property 1 (Scalar Inverse). *For $\lambda \neq 0$,*

$$(\lambda \mathbf{A})^{-1} = \frac{1}{\lambda} \mathbf{A}^{-1}.$$

Demonstration:

$$\begin{aligned} (\lambda \mathbf{A}) \left(\frac{1}{\lambda} \mathbf{A}^{-1} \right) &= \mathbf{A} \mathbf{A}^{-1} = \mathbf{I}, \\ \left(\frac{1}{\lambda} \mathbf{A}^{-1} \right) (\lambda \mathbf{A}) &= \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}. \end{aligned}$$

Property 2 (Inverse of a Product).

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

Demonstration:

$$\begin{aligned} (\mathbf{AB})(\mathbf{B}^{-1} \mathbf{A}^{-1}) &= \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} = \mathbf{AA}^{-1} = \mathbf{I}, \\ (\mathbf{B}^{-1} \mathbf{A}^{-1})(\mathbf{AB}) &= \mathbf{B}^{-1}(\mathbf{A}^{-1} \mathbf{A})\mathbf{B} = \mathbf{B}^{-1} \mathbf{B} = \mathbf{I}. \end{aligned}$$

Property 3 (Inverse Transpose).

$$(\mathbf{A}^{-1})' = (\mathbf{A}')^{-1}.$$

Demonstration:

$$\begin{aligned} \mathbf{A}'(\mathbf{A}^{-1})' &= (\mathbf{A}^{-1} \mathbf{A})' = \mathbf{I}' = \mathbf{I}, \\ (\mathbf{A}^{-1})' \mathbf{A}' &= (\mathbf{A} \mathbf{A}^{-1})' = \mathbf{I}' = \mathbf{I}. \end{aligned}$$

Property 4 (Sum Inverse).

$$(\mathbf{A} + \mathbf{B})^{-1} = \mathbf{A}^{-1}(\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1} \mathbf{B}^{-1}.$$

Demonstration: Factor $\mathbf{A} + \mathbf{B}$ by inserting identity matrices:

$$\mathbf{A} + \mathbf{B} = \mathbf{BB}^{-1} \mathbf{A} + \mathbf{BA}^{-1} \mathbf{A} = \mathbf{B}(\mathbf{B}^{-1} + \mathbf{A}^{-1})\mathbf{A}.$$

Inverting both sides using Property 2:

$$(\mathbf{A} + \mathbf{B})^{-1} = (\mathbf{B}(\mathbf{A}^{-1} + \mathbf{B}^{-1})\mathbf{A})^{-1} = \mathbf{A}^{-1}(\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1} \mathbf{B}^{-1}.$$

Property 5 (Inverse Difference).

$$\mathbf{A}^{-1} - (\mathbf{A} + \mathbf{B})^{-1} = \mathbf{A}^{-1}(\mathbf{B}^{-1} + \mathbf{A}^{-1})^{-1} \mathbf{A}^{-1}.$$

Demonstration: Factor $\mathbf{A} + \mathbf{B}$ differently:

$$\mathbf{A} + \mathbf{B} = \mathbf{A}\mathbf{A}^{-1}\mathbf{B} + \mathbf{A}\mathbf{B}^{-1}\mathbf{B} = \mathbf{A}(\mathbf{A}^{-1} + \mathbf{B}^{-1})\mathbf{B}.$$

So $(\mathbf{A} + \mathbf{B})^{-1} = \mathbf{B}^{-1}(\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1}\mathbf{A}^{-1}$. Now write $\mathbf{B}^{-1} = (\mathbf{B}^{-1} + \mathbf{A}^{-1}) - \mathbf{A}^{-1}$ and expand:

$$\begin{aligned} (\mathbf{A} + \mathbf{B})^{-1} &= [(\mathbf{B}^{-1} + \mathbf{A}^{-1}) - \mathbf{A}^{-1}](\mathbf{B}^{-1} + \mathbf{A}^{-1})^{-1}\mathbf{A}^{-1} \\ &= \mathbf{I} \cdot \mathbf{A}^{-1} - \mathbf{A}^{-1}(\mathbf{B}^{-1} + \mathbf{A}^{-1})^{-1}\mathbf{A}^{-1} \\ &= \mathbf{A}^{-1} - \mathbf{A}^{-1}(\mathbf{B}^{-1} + \mathbf{A}^{-1})^{-1}\mathbf{A}^{-1}. \end{aligned}$$

Rearranging gives the result.

Structural Properties

Property 6 (Idempotent Matrices). *If $\mathbf{H}$ is a $k \times k$ idempotent matrix ($\mathbf{H}^2 = \mathbf{H}$), then $\mathbf{H}$ is either singular or the identity matrix, and $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H})$.*

Demonstration (singular or identity): If $\mathbf{H}$ is nonsingular, multiply $\mathbf{H}^2 = \mathbf{H}$ on the left by $\mathbf{H}^{-1}$:

$$\mathbf{H}^{-1}\mathbf{H}^2 = \mathbf{H}^{-1}\mathbf{H} \implies \mathbf{H} = \mathbf{I}.$$

So the only nonsingular idempotent matrix is $\mathbf{I}$.

Demonstration (trace equals rank): Since $\mathbf{H}^2 = \mathbf{H}$, the eigenvalues of $\mathbf{H}$ must satisfy $\lambda^2 = \lambda$, so each eigenvalue is 0 or 1. The number of eigenvalues equal to 1 is $\text{rank}(\mathbf{H})$, and the trace is the sum of all eigenvalues, so $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H})$.

Example: The OLS projection matrix $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is idempotent:

$$\begin{aligned} \mathbf{P}^2 &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{P}. \end{aligned}$$

Note also that $\mathbf{P}' = \mathbf{X}((\mathbf{X}'\mathbf{X})^{-1})'\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{P}$, since $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric. So $\mathbf{P}$ is both idempotent and symmetric.

Property 7 (Orthogonal Matrices). *If a $k \times k$ matrix $\mathbf{H}$ satisfies $\mathbf{H}'\mathbf{H} = \mathbf{I}$, then:*

1. $\mathbf{H}$ is nonsingular with $\mathbf{H}^{-1} = \mathbf{H}'$,
2. $\mathbf{H}\mathbf{H}' = \mathbf{I}$,
3. $(\mathbf{H}')^{-1} = \mathbf{H}$.

Demonstration:

(1) The equation $\mathbf{H}'\mathbf{H} = \mathbf{I}$ says that $\mathbf{H}'$ is a left inverse of $\mathbf{H}$. Taking determinants: $\det(\mathbf{H}')^2 = \det(\mathbf{H})^2 = 1$, so $\det(\mathbf{H}) \neq 0$ and $\mathbf{H}$ is nonsingular. Since the left inverse of a nonsingular matrix equals its inverse, $\mathbf{H}^{-1} = \mathbf{H}'$.

(2) Multiply $\mathbf{H}'\mathbf{H} = \mathbf{I}$ on the left by $\mathbf{H}$ and on the right by $\mathbf{H}^{-1} = \mathbf{H}'$:

$$\mathbf{H}\mathbf{H}'\mathbf{H}\mathbf{H}' = \mathbf{H}\mathbf{H}' \implies (\mathbf{H}\mathbf{H}')^2 = \mathbf{H}\mathbf{H}'.$$

But more directly: $\mathbf{H}\mathbf{H}' = \mathbf{H}\mathbf{H}^{-1} = \mathbf{I}$.

(3) From (1), $\mathbf{H}^{-1} = \mathbf{H}'$. By the inverse transpose property (Property 3), $(\mathbf{H}')^{-1} = (\mathbf{H}^{-1})^{-1} = \mathbf{H}$.

Matrix Inversion Lemmas

The following identities are useful for updating inverses when a matrix is perturbed. Each is demonstrated by verifying that the proposed inverse, when multiplied by the original matrix, yields I .

Property 8 (Woodbury Matrix Identity). *For nonsingular $n \times n$ matrix A , $n \times k$ matrix B , nonsingular $k \times k$ matrix C , and $k \times n$ matrix D :*

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}.$$

Demonstration: Let $R = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}$. We verify $(A + BCD)R = I$. Write $S = C^{-1} + DA^{-1}B$ for brevity.

$$\begin{aligned} (A + BCD)R &= I - BS^{-1}DA^{-1} + BCDA^{-1} - BCDA^{-1}BS^{-1}DA^{-1} \\ &= I + B[CDA^{-1} - S^{-1}DA^{-1} - CDA^{-1}BS^{-1}DA^{-1}] \\ &= I + B[CDA^{-1} - (I + CDA^{-1}B)S^{-1}DA^{-1}]. \end{aligned}$$

Now $I + CDA^{-1}B = C(C^{-1} + DA^{-1}B) = CS$, so:

$$\begin{aligned} &= I + B[CDA^{-1} - CSS^{-1}DA^{-1}] \\ &= I + B[CDA^{-1} - CDA^{-1}] = I. \quad \square \end{aligned}$$

Property 9 (Sherman–Morrison Formula). *For nonsingular A and column vector b , provided $1 + b'A^{-1}b \neq 0$:*

$$(A + bb')^{-1} = A^{-1} - \frac{A^{-1}bb'A^{-1}}{1 + b'A^{-1}b}.$$

Demonstration: This is the Woodbury identity with $B = b$, $C = 1$, $D = b'$:

$$(A + b \cdot 1 \cdot b')^{-1} = A^{-1} - A^{-1}b(1 + b'A^{-1}b)^{-1}b'A^{-1} = A^{-1} - \frac{A^{-1}bb'A^{-1}}{1 + b'A^{-1}b}.$$

The subtraction variant follows from Woodbury with $B = b$, $C = -1$, $D = b'$:

$$(A - bb')^{-1} = A^{-1} - A^{-1}b(-1 + b'A^{-1}b)^{-1}b'A^{-1} = A^{-1} + \frac{A^{-1}bb'A^{-1}}{1 - b'A^{-1}b},$$

provided $1 - b'A^{-1}b \neq 0$.